Homework 28: 10.10-10.12 **Name** _____

AP Calculus BC **Due** _____

Period _____ **Score** _____ **/ 100**

Skill Builder 10.10 – Alternating Series Error Bound

Approximate the sum of each series by using the first five terms. Write your approximation as a compound inequality to six decimal places. You may use your calculator.

1.) $\sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)!}$	**2.)** $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$
3.) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$	**4.)** $\sum_{k=1}^{\infty} (-1)^k \frac{1}{2k+1}\left(\frac{1}{3}\right)^{2k+1}$

Determine the number of terms required to approximate the sum of the series with an error less than 0.001.

5.) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$

6.) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n^3 - 1}$

7.) $\sum_{n=1}^{\infty} \frac{(-1)^{n}}{n!}$

Topic 10.11 – Finding Taylor Polynomial Approximations of a Function

For the following, find the Maclaurin polynomial of degree *n* for the given function and evaluate at $x = 0.1$.

1.) $f(x) = e^{-x},\ n = 5$	**2.)** $f(x) = \dfrac{1}{1-x},\ n = 5$
3.) $f(x) = \sin(\pi x),\ n = 3$	**4.)** $f(x) = xe^{x},\ n = 3$
5.) $f(x) = \tan x,\ n = 3$	**6.)** $f(x) = \tan^{-1} x,\ n = 3$

For the following, find the Taylor polynomial of degree n for the given function centered at $x = c$.

7.) $f(x) = \frac{1}{x},\ n = 4, c = 1$	**8.)** $f(x) = x\cos x,\ n = 2, c = \pi$

Topic 10.12 – Lagrange Error Bound

1.) Estimate the error in approximating $\sin(0.5)$ by the fifth-degree polynomial by finding the Lagrange remainder.

2.) A student needs to calculate $\frac{1}{\sqrt{e}}$ on a four-function calculator. He uses a third-degree Taylor polynomial for $y = e^x$ and adds on the Lagrange error. What is the result of his calculation?

3.) Let f be a function given by $f(x) = \cos\left(3x - \frac{\pi}{10}\right)$ and let $P(x)$ be the fourth-degree Taylor polynomial about f at $x = 0$. Use the Lagrange error bound to show that $\left| f\left(\frac{1}{4}\right) - P\left(\frac{1}{4}\right) \right| < \frac{1}{500}$.

4.) Let f be a function having derivatives for all orders of real numbers. The first three derivatives of f at $x=2$ are given in the table below. The fourth derivative of f satisfies the inequality $\left|f^{(4)}(x)\right| \le 50$ for all x in the interval $[2, 2.5]$. Use the Lagrange error bound on this approximation to find $f(2.1)$ to explain why $f(2.1) < 8.5$.

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
2	8	4	16	18

Taylor Polynomials and Error Bounds on the AP Exam – Circuit Training

Directions: Beginning in the first cell, find the answer. Hunt for that answer elsewhere in the document, mark that cell #2 and work the problem in that box. Process in this manner until you complete the circuit. Show all pertinent work. Calculators may be used.

__1__ Answer: $\frac{5}{192}$

Let f be a function having derivatives for all orders of real numbers. The first three derivatives of f at $x=0$ are given in the table below. Use the third-degree Taylor polynomial at $x=0$ to approximate $f\left(\frac{1}{2}\right)$.

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	–2	2	3	–4

____ Answer: $\frac{25}{4}$

Let f be a function having derivatives for all orders of real numbers. The 4th degree Taylor polynomial for f about $x = -3$ is given by $P_4(x) = -6 - \frac{2}{7}(x+3)^2 + k(x+3)^4$, where $k \ne 0$. If $f^{(4)}(-3) = \frac{3\pi}{4}$, find k.

# ____ Answer: 0.225 A function has derivatives of all orders at $x = 0$. Let $P_n(x)$ denote the nth degree Taylor polynomial about $x = 0$. It is known that $f(0) = 4$, $P_1\left(\frac{1}{2}\right) = -2$, and $P_2\left(\frac{1}{2}\right) = -6$. Find $f''(0)$.	# ____ Answer: $\frac{25}{32}$ Find the error bound for the 2nd degree Taylor polynomial of $f(x) = \cos x$ centered at $x = 0$ on $\left[0, \frac{\pi}{2}\right]$.
# ____ Answer: $\frac{5}{16}$ The 3rd degree Taylor polynomial for f about $x = 2$ is given by the following: $P_3(x) = -5(x-2) + \frac{7}{2}(x-2)^2 + \frac{10}{3}(x-2)^3$. The 4th derivative of f satisfies the inequality $\left\lvert f^{(4)}(x)\right\rvert \le 10$ for all x in the interval [2, 4]. Find the Lagrange error bound for the approximation to $f(2.5)$.	# ____ Answer: $\frac{\pi}{32}$ What is the coefficient for x^3 in the Taylor polynomial for $\frac{1}{\sqrt{1-x}}$ about $x = 0$?

____ Answer: $\frac{\pi^4}{384}$

Given the 5th degree Taylor polynomial, $P_5(x) = 1 - 2(x-1) + 4(x-1)^2 - 3(x-1)^4 - \frac{2}{3}(x-1)^5$, for the function f about $x = 1$. What is the value of $f^{(5)}(1)$?

____ Answer: $-\frac{17}{24}$

Let f be a function having derivatives of all orders for $x > 0$. Selected values of f and its first three derivatives are indicated in the table below. The function f and these three derivatives are increasing on the interval $0 \le x \le 2$. Find the Lagrange error bound for $f(0.7)$ using the 2nd degree Taylor polynomial for f about $x = 1$.

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	5	3	14	20
1	9	11	33	50
2	16	24	56	76

____ Answer: –32

Use the 2nd degree Taylor polynomial about $x = 0$ for $f(x) = e^{-x/2}$ to approximate $e^{-1/4}$.

____ Answer: –80

Let f be a function having derivatives for all orders of real numbers. The function and its first four derivatives at $x = 0$ are given in the table below. The 5th derivative of f satisfies the inequality $\left|f^{(5)}(x)\right| \le 150$ for all x in the interval [0, 2]. Find the maximum value of $f(1)$.

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(4)}(x)$
0	2	3	0	5	–60